

Martin Špetlík

AI RESEARCHER · SCIENTIFIC COMPUTING · HPC

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Personal Profile

Applied machine learning researcher and software engineer with a PhD in Applied Sciences in Engineering, specialising in deep learning, uncertainty quantification, and high-performance computing (HPC). Experienced in designing and training CNN, GNN, and generative models (VQ-VAE, VQ-GAN, diffusion) at scale on HPC systems (Slurm, PBS, GPU). Lead developer of the open-source **mlmc** Python package (PyPI). Proven track record of translating complex scientific problems into reproducible and scalable machine learning workflows, including peer-reviewed publications, participation in European research projects, and close collaboration with domain scientists to apply AI in practice.

Work Experience

Research Assistant

Liberec, Czech Republic

Technical University of Liberec

Sep 2023 – Present

- Led machine learning strategy for a generative AI project on 3D CT scans of pelvic bones; designed VQ-GAN and diffusion-based pipelines for high-resolution volumetric imaging data under low-sample conditions.
- Trained and evaluated models on HPC clusters (Slurm, PBS); optimised job scheduling, GPU memory usage, and runtime performance while maintaining reproducible experimental workflows.
- Integrated an Unscented Kalman Filter (UKF) with a nonlinear Richards-equation model for data assimilation; containerised the full workflow (Docker, Singularity) for reproducibility.
- Lead developer and maintainer of the open-source **mlmc** Python package (<https://pypi.org/project/mlmc/>), v1.0.3 (Oct 2025), supporting parallel execution and HDF5-based data management on HPC clusters.
- Designed and delivered course materials for Computer Architecture and contributed seminar content for Graph Theory.

Research Intern

Mol, Belgium

Belgian Nuclear Research Center (SCK CEN)

Feb 2023 – Jul 2023

- Designed and implemented a deep-learning surrogate model for numerical homogenisation in heterogeneous rock environments, including dataset construction, model design, validation, and HPC training.
- Contributed to European nuclear waste repository safety research within the EURAD framework.

Junior Researcher

Liberec, Czech Republic

Technical University of Liberec

Jan 2020 – Dec 2022

- Introduced graph neural networks (GNNs) into the research group to accelerate large-scale physical simulation workflows.
- Developed ML-based surrogate models within Multilevel Monte Carlo (MLMC) frameworks to improve computational efficiency for uncertainty quantification.

Python developer (part-time)

Liberec, Czech Republic

Technical University of Liberec

Jul 2018 – Dec 2019

- Developed the initial version of the open-source **mlmc** Python package, establishing its core architecture and feature set.

Back-End Developer (part-time)

Liberec, Czech Republic

Hotel.cz

Jan 2017 – Jun 2018

- Developed and maintained production back-end services using PHP, SQL, and Linux-based infrastructure.

Education

Ph.D. in Applied Sciences in Engineering

Liberec, Czech Republic

Technical University of Liberec

Sep 2020 – Apr 2026

- Thesis: *Multilevel Monte Carlo Method for Models on Mixed Meshes*. Defended April 2026.
- Supervisor: Assoc. Prof. Dr. Jan Březina
- First application of Multilevel Monte Carlo (MLMC) method to stochastic 3D discrete fracture-matrix (DFM) models; developed 3D CNN surrogate models achieving confirmed **100×** speedup over conventional numerical homogenisation and **>90%** reduction in computational cost vs. standard Monte Carlo method.

GeMSS - Generative Modeling Summer School

Eindhoven, The Netherlands

Eindhoven University of Technology

June 2024

- Completed intensive 1-week training focused on generative models including VAEs, GANs, diffusion models, or transformer-based architectures.

Master's Degree in Information Technology

Liberec, Czech Republic

Technical University of Liberec

July 2020

- Thesis: *Reconstruction of Distribution from Estimated Generalized Moments*.
- Graduated Cum Laude; recipient of the Dean's Award for Outstanding Master's Thesis.

Key Research Projects

- Generative AI for 3D CT Scans**

Liberec, Czech Republic
- Czech Science Foundation, Grant No. 24-10862S

2024 - Present
- Led the machine learning strategy and architectural design for generative modeling of 3D CT scans of pelvic bones.
 - Developed a specialized pipeline combining VQ-GAN and Denoising Diffusion Models to address challenges of low-sample, high-resolution medical-imaging data.
 - Collaboration with the anatomy department: expert-guided assessment of generated 3D pelvic CT volumes, with feedback on anatomical plausibility.
- Multilevel Monte Carlo (MLMC) Acceleration via Deep-Learning Surrogates**

Liberec, Czech Republic
- SGS-2025-3560, SGS-2023-3379; EURAD-2 (Grant Agreement ID 101166718)

2023 - Present
- Principal Investigator of two student grants focused on ML-based acceleration of stochastic simulations; Research Team Member for European and National research grants.
 - Designed and deployed convolutional surrogate models to replace computationally expensive numerical homogenization in 2D and 3D fracture-matrix systems.
- Data Assimilation in Non-Linear Models**

Liberec, Czech Republic
- Technology Agency of the Czech Republic, SS06010280

2023 - Present
- Responsible for data assimilation using Unscented Kalman Filters (UKF) in a nonlinear Richards-equation model.
 - Designed experimental setups and integration strategies for UKF-based approaches applied to nonlinear simulation models.
- Open-Source Software Development, UQ, and GNN Integration**

Liberec, Czech Republic
- Technology Agency of the Czech Republic (TH03010227, TK02010118); EURAD (Grant Agreement ID 847593)

2018 - Present
- Lead contributor to the open-source `mlmc` Python package, responsible for its architecture, development, and integration with external tools.
 - Applied Uncertainty Quantification (UQ) methods to groundwater processes and introduced Graph Neural Networks (GNNs) as surrogates for accelerating UQ analyses.
 - Ensured software portability, reproducibility, and maintainability for long-term multi-partner scientific projects.

Skills

Deep Learning	Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), Generative AI (VAEs, GANs, Diffusion Models), Vector Quantization (VQ-VAE, VQ-GAN), Vision Transformer, Representation learning for 3D and geometric data
Scientific Computing	Multilevel Monte Carlo (MLMC), Surrogate Modeling, Uncertainty Quantification (UQ), Kalman Filters (including Unscented KF), High-Dimensional Simulation Modeling, High-Performance Computing (HPC; job scheduling - PBS, Slurm, GPU-based training)
Programming & Tools	Python (PyTorch, TensorFlow, Spektral, NumPy, Pandas, Scikit-learn), MLflow, Git, Linux, Docker, Singularity, HDF5, Zarr, Xarray, SQL, Dask, $\LaTeX$
Research & Development	Reproducible research, Experimental design, Model development and evaluation, Workflow automation, Technical documentation, Independent problem-solving, Literature-driven method development
Soft Skills	Team collaboration, Critical thinking, Technical communication, Time management, Teaching and mentoring

Publications

DISSERTATION

Multilevel Monte Carlo Method for Models on Mixed Meshes

Technical University of Liberec

2026

- PhD dissertation in Applied Sciences in Engineering

JOURNAL ARTICLES

Convolutional Surrogate for 3D Discrete Fracture-Matrix Tensor Upscaling

Martin Špetlík, Jan Březina

Computers and Geosciences. 2026, URL: <https://www.sciencedirect.com/science/article/pii/S0098300426000026>

Deep learning surrogate for predicting hydraulic conductivity tensors from stochastic discrete fracture-matrix models

Martin Špetlík, Jan Březina, Eric Laloy

Computational Geosciences. 2024, URL: <https://link.springer.com/article/10.1007/s10596-024-10324-8>

Groundwater Contaminant Transport Solved by Monte Carlo Methods Accelerated by Deep Learning Meta-Model

Martin Špetlík, Jan Březina

Applied Sciences. 2022, URL: <https://www.mdpi.com/2076-3417/12/15/7382>

CONFERENCE PROCEEDINGS

Groundwater flow meta-model for multilevel Monte Carlo methods

Martin Špetlík, Jan Březina

CEUR Workshop Proceedings, 2021, URL: <https://ceur-ws.org/Vol-2962/paper40.pdf>

Languages

Czech	Native
English	Full Professional Proficiency
German	Professional working proficiency